Inertial & Gravitational mass is the same.

compared to Coulomb's Law: $F_c \propto \frac{q_1 q_2}{r^2}\hat{r}$

$$\vec{F} = m\vec{a}$$

same

gravitational charges.

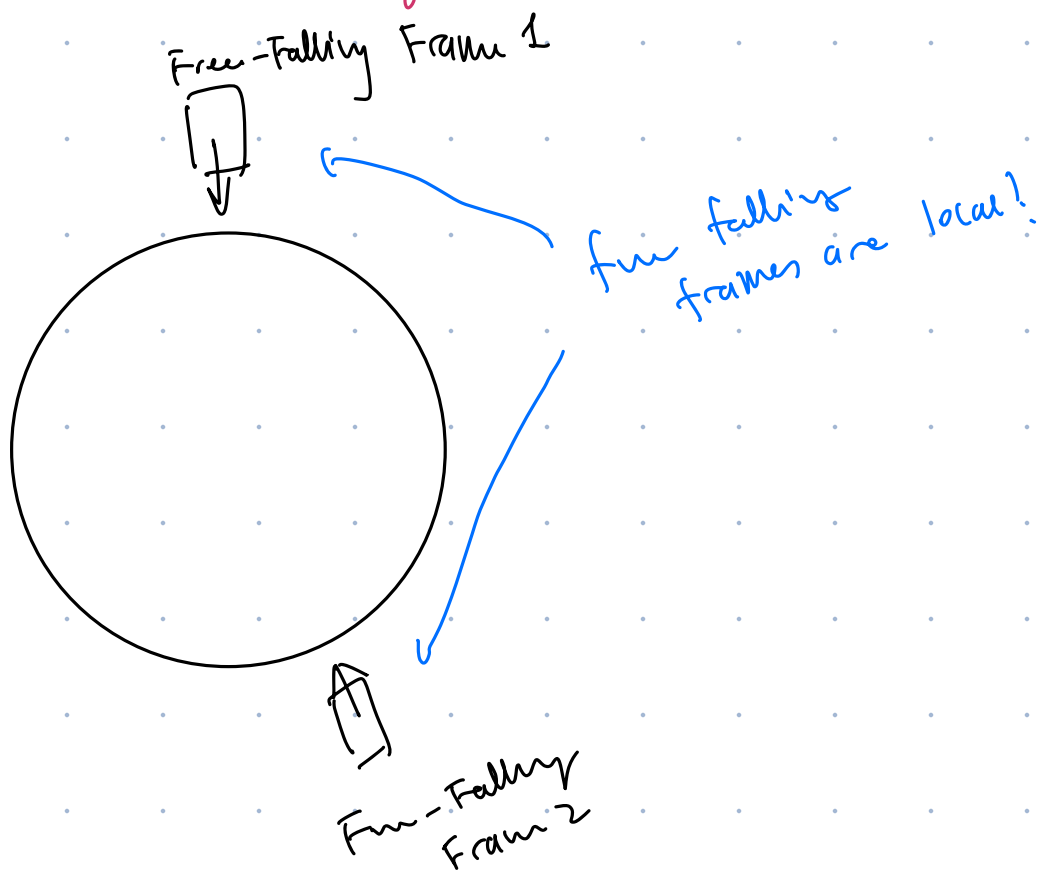
$$\vec{F} = -G_N \frac{m_1 m_2}{r^2}\hat{r}$$

Einstein's Insight: There is no way you can distinguish acceleration from gravity.

↳ when you go to a free-falling frame

↳ No gravity

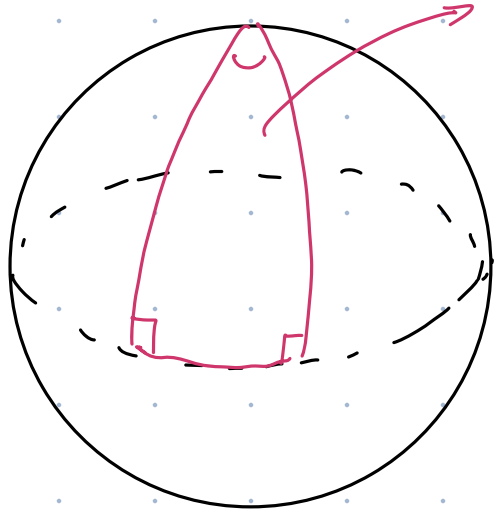
↳ SR rules apply



3rd Part of the Class:

→ How to transform from 1 local frame → another local frame

→

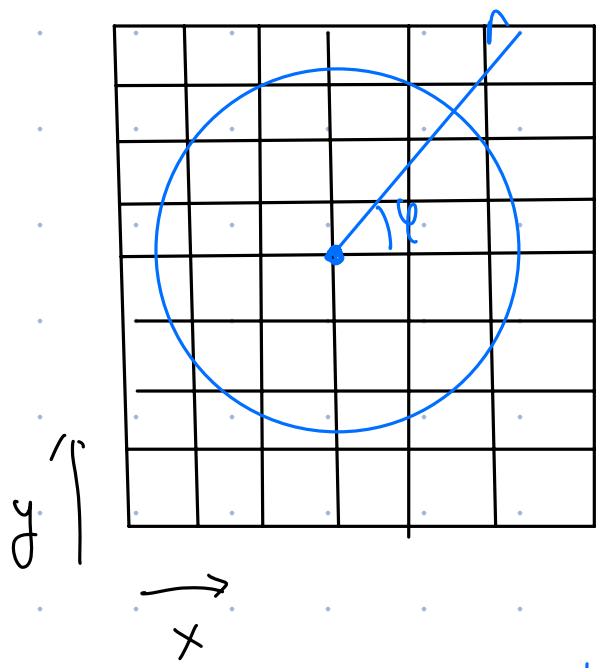


sum of angles > 180°

positive curvature.

For an arbitrary coordinate (σ, τ), there has to be a unique mapping from $(\sigma, \tau) \to (x, y)$

→



$\sigma(x,y)$

$\tau(x,y)$

$$\Delta\sigma = \frac{d\sigma}{dx}\Delta x + \frac{d\sigma}{dy}\Delta y$$

$$\Delta\tau = \frac{d\tau}{dx}\Delta x + \frac{d\tau}{dy}\Delta y$$

$$\det\begin{pmatrix} \frac{d\sigma}{dx} & \frac{d\sigma}{dy} \\ \frac{d\tau}{dx} & \frac{d\tau}{dy} \end{pmatrix} \neq 0$$

except perhaps at certain points

<u>For example:</u>

$$\begin{cases} r = \sqrt{x^2+y^2} \\ \varphi = \tan^{-1}\left(\frac{y}{x}\right) \end{cases} \Leftrightarrow \begin{cases} x = r\cos\theta \\ y = r\sin\theta \end{cases}$$

$$\Lambda^{\alpha'}{}_{\alpha} = \frac{\partial x^{\alpha'}}{\partial x^{\alpha}} \qquad \alpha' = 1,2 \quad (r,\varphi) \text{ or } (\sigma, 2)$$

$$\alpha = 1,2 \quad (x,y)$$

↳ primed indices ⇒ one coordinate system,

↳ unprimed indices ⇒ another coordinate system.

Some important quantities.

$$\frac{\partial r}{\partial x} = \frac{2x}{2\sqrt{x^2+y^2}} = \frac{x}{\sqrt{x^2+y^2}} = \cos\varphi$$

$$\frac{\partial r}{\partial y} = \frac{y}{\sqrt{x^2+y^2}} = \sin\varphi$$

$$\frac{\partial \varphi}{\partial x} = \frac{-y/x^2}{1+\frac{y^2}{x^2}} = \frac{-y}{x^2+y^2} = -\frac{\sin\varphi}{r}$$

$$\frac{\partial \varphi}{\partial y} = \frac{\frac{1}{x}}{1+\frac{y^2}{x^2}} = \frac{x}{x^2+y^2} = \frac{\cos\varphi}{r}$$

$$\frac{\partial x}{\partial r} = \cos\varphi, \quad \frac{\partial y}{\partial r} = \sin\varphi, \quad \frac{dx}{d\varphi} = -r\sin\varphi,$$

$$\frac{dy}{d\varphi} = r\cos\varphi$$

$$\Lambda^r{}_x = \frac{\partial r}{\partial x} = \cos\varphi$$

$$\Lambda^{\alpha'}{}_{\alpha} = \begin{array}{c} \begin{array}{cc} \Lambda^0{}_0 & \Lambda^0{}_1 \end{array} \\ \begin{pmatrix} \cos\varphi & \sin\varphi \\ -\frac{\sin\varphi}{r} & \frac{\cos\varphi}{r} \end{pmatrix} \\ \begin{array}{cc} \Lambda^1{}_0 & \Lambda^1{}_1 \end{array} \end{array} \qquad \Lambda^{\alpha}{}_{\alpha'} = \begin{array}{c} \begin{array}{cc} \Lambda^0{}_0 & \Lambda^0{}_1 \\ \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \varphi} \end{array} \\ \begin{pmatrix} \cos\varphi & -r\sin\varphi \\ \sin\varphi & r\cos\varphi \end{pmatrix} \\ \begin{array}{cc} \Lambda^1{}_0 & \Lambda^1{}_1 \end{array} \end{array}$$

$$\Lambda^{\alpha'}{}_{\alpha}\Lambda^{\alpha}{}_{\alpha'} = \begin{pmatrix} c & s \\ -\frac{s}{r} & \frac{c}{r} \end{pmatrix} \begin{pmatrix} c & -rs \\ s & rc \end{pmatrix}$$

$$= \begin{pmatrix} c^2+s^2 & -rsc+src \\ -\frac{s}{r}c+\frac{cs}{r} & +\frac{rs\cdot s}{r}+\frac{c}{r}rc \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

these coordinate transformations are space-dependent. I.O.W they are local.

$$t' = \gamma(t + vx)$$

$$x' = \gamma(x + vt)$$

$$\Lambda^{t'}{}_{t} = \frac{\partial t'}{\partial t} = \gamma$$

$$\Lambda^{t'}{}_{x} = \gamma v$$

$$\Lambda^{x'}{}_{x} = \gamma$$

$$\Lambda^{x'}{}_{t} = \gamma v$$

$$\Lambda^{\alpha'}{}_{\alpha} = \begin{pmatrix} \gamma & \gamma v \\ \gamma v & \gamma \end{pmatrix}$$

this is global

(does not depend on space & time).

$$\underbrace{V^{\alpha'} = \Lambda^{\alpha'}{}_{\alpha} V^{\alpha}}_{\text{a vector}}$$

↳ transforms the same way as infinitesimal coordinate differences.

↳ A tensor

$$T^{\alpha'\beta'\gamma'} = \Lambda^{\alpha'}{}_{\alpha} \Lambda^{\beta'}{}_{\beta} \Lambda^{\gamma'}{}_{\gamma} T^{\alpha\beta\gamma}$$

Basis for Vectors

$$\hat{x} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \hat{y} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\updownarrow \qquad\qquad \updownarrow$$

$$e_{(\hat{x})} \qquad e_{(\hat{y})} \qquad e_{(\alpha)}$$

$$V = V^{\alpha} e_{\alpha} = V^{1} e_{(1)} + V^{2} e_{(2)}$$

$$= V^{\alpha'} e_{(\alpha')}$$

Coordinate independent.

$$V = V^{\alpha'} e_{(\alpha')} = \boxed{\Lambda^{\alpha'}{}_{\alpha} \boxed{V^{\alpha}} e_{(\alpha')}} = V^{\alpha} e_{(\alpha)}$$

$$e_{(\alpha)} = \Lambda^{\alpha'}{}_{\alpha} e_{\alpha'}$$

should transform like this:

this is smart!

Multiply by $\Lambda^{\alpha}{}_{\beta'} \Rightarrow$

$$\boxed{e_{(\alpha)} \Lambda^{\alpha}{}_{(\beta')}} = \underbrace{\Lambda^{\alpha'}{}_{\alpha} \Lambda^{\alpha}{}_{\beta'}}_{\delta^{\alpha'}{}_{\beta'}} e_{(\alpha')}$$

$$\delta^{\alpha'}{}_{\beta'} e_{(\alpha')} = \boxed{e_{\beta'}}$$

$\Downarrow$

For example:

$$\underbrace{e_{(2')} = \Lambda^{1}{}_{2'}\, e_{(1)} + \Lambda^{2}{}_{(2')}\, e_{(2)}}$$

$\downarrow$

this tells you how to represent one basis vector in another basis

↳ Why do coordinates transform in the opposite way?

↳ any intuition?

→ a vector is an invariant quantity:

$$V = V^{\alpha} e_{\alpha} = V^{\alpha'} e_{\alpha'}$$

$$= \Lambda^{\alpha}{}_{\alpha'} V^{\alpha'} \Lambda e_{\alpha'}$$

$$\Lambda^{\alpha}{}_{\alpha'} \Lambda^{\alpha'}{}_{\alpha} = I.$$

choose this s.t.

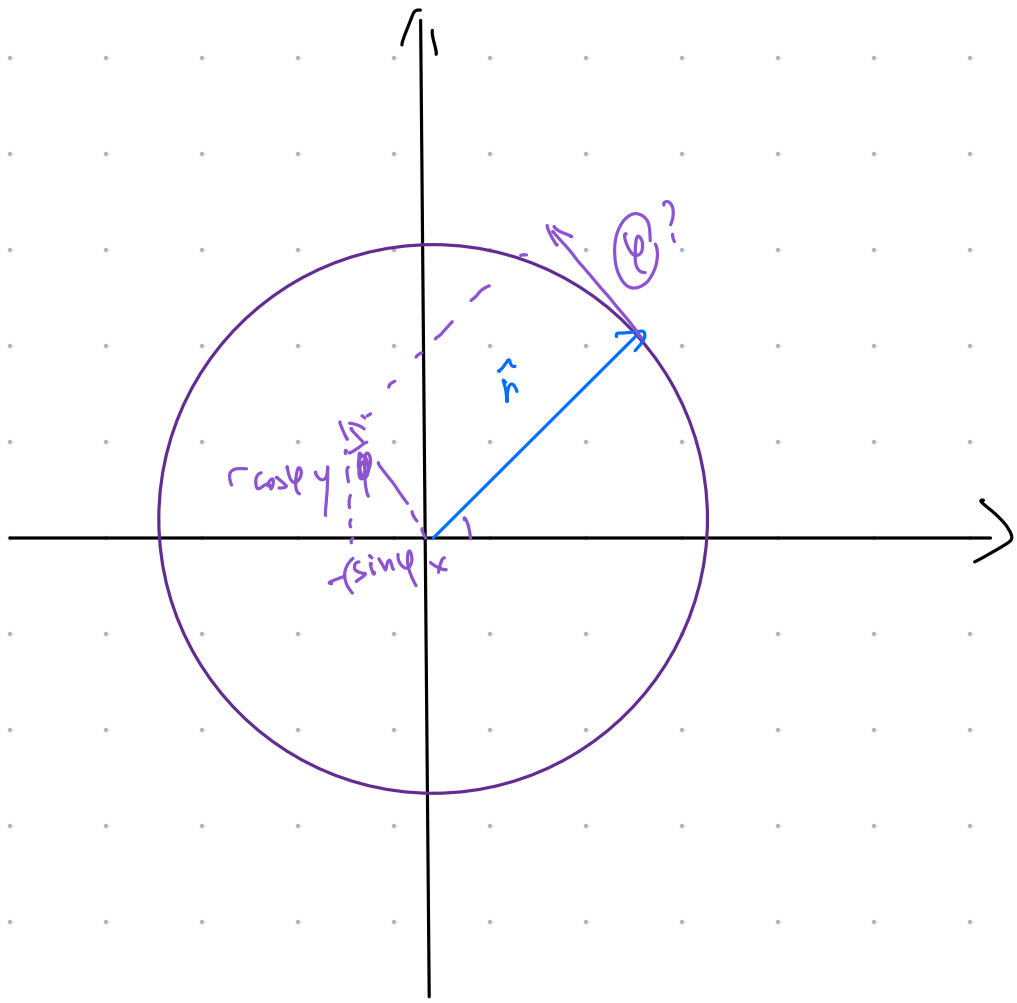
$$e_{(r)} = \Lambda^{\alpha}{}_{(r)}\, e_{\alpha} = \Lambda^{x}{}_{r}\, e_x + \Lambda^{y}{}_{r}\, e_y$$

$$= \cos\varphi\, e_x + \sin\varphi\, e_y$$

$$= \cos\varphi\, \hat{x} + \sin\varphi\, \hat{y}.$$

$$e_{(\varphi)} = \Lambda^{\alpha}{}_{(\varphi)}\, e_{\alpha} = \Lambda^{x}{}_{\varphi}\, e_x + \Lambda^{y}{}_{\varphi}\, e_y$$

$$= -r\sin\varphi\, \hat{x} + r\cos\varphi\, \hat{y}.$$



$$e(\varphi) = \frac{d}{d\varphi}(r\cos\varphi,\ r\sin\varphi)$$

$$= r(-\sin\varphi,\ \cos\varphi)$$

$e_x \cdot e_x = 1$

$e_y \cdot e_y = 1$

$e_x \cdot e_y = 0$

$e_{(r)} \cdot e_{(r)} = \cos^2\varphi + \sin^2\varphi = 1$

$e_{(r)} \cdot e_{(\varphi)} = -r\cos\varphi\sin\varphi + r\cos\varphi\sin\varphi = 0$

$$\boxed{e_{(\varphi)} \cdot e_{(\varphi)} = r^2}$$